Tabular Q-learning on FrozenLake

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Public code repository: https://github.com/RohitVummadi/CPSC-4420-Assignment2

A. Environment

The agent moves from the upper-left starting tile to the lower-right goal on the slippery 8 by 8 FrozenLake map. It must avoid holes. The observation space contains 64 discrete tile states, numbered 0 through 63. A tile at zero-based row r and column c has state s = 8r + c. No discretization is needed.

The four actions are 0 = left, 1 = down, 2 = right, and 3 = up. With slippery movement enabled, the intended direction and each perpendicular direction have probability 1/3. A move against the boundary leaves the agent in place. Reaching the goal gives reward +1; holes and all other moves give 0. A hole or the goal terminates an episode. The installed environment truncates an episode after 100 steps.

B. Q-learning method

I implemented a 64 by 4 Q-table initialized to zero. Epsilon-greedy action selection chooses a random action with probability epsilon; otherwise it chooses among the actions with the highest Q-value, breaking training ties randomly. Each transition updates the value of the intended state-action pair:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\left[r + \gamma \max_{a\prime} Q(s\prime, a\prime) - Q(s,a)\right]$$

For a terminal transition, the target is the observed reward only. For other transitions, it includes the discounted next-state value. Truncation ends the training loop but does not remove this next-state term. Epsilon decays after each episode and stops at its minimum. Each new training run resets its Q-table, epsilon, and reward history. The learning rate, discount factor, exploration schedule, and episode count are easy to change in the notebook.

C. Baseline training

Setting	Value
Environment	FrozenLake-v1, 8 by 8, slippery, 100-step limit
Learning rate / discount factor	alpha = 0.1 / gamma = 0.99
Exploration schedule	epsilon starts at 1.0; multiply by 0.9995 each episode
Minimum epsilon / episodes	0.05 / 20,000

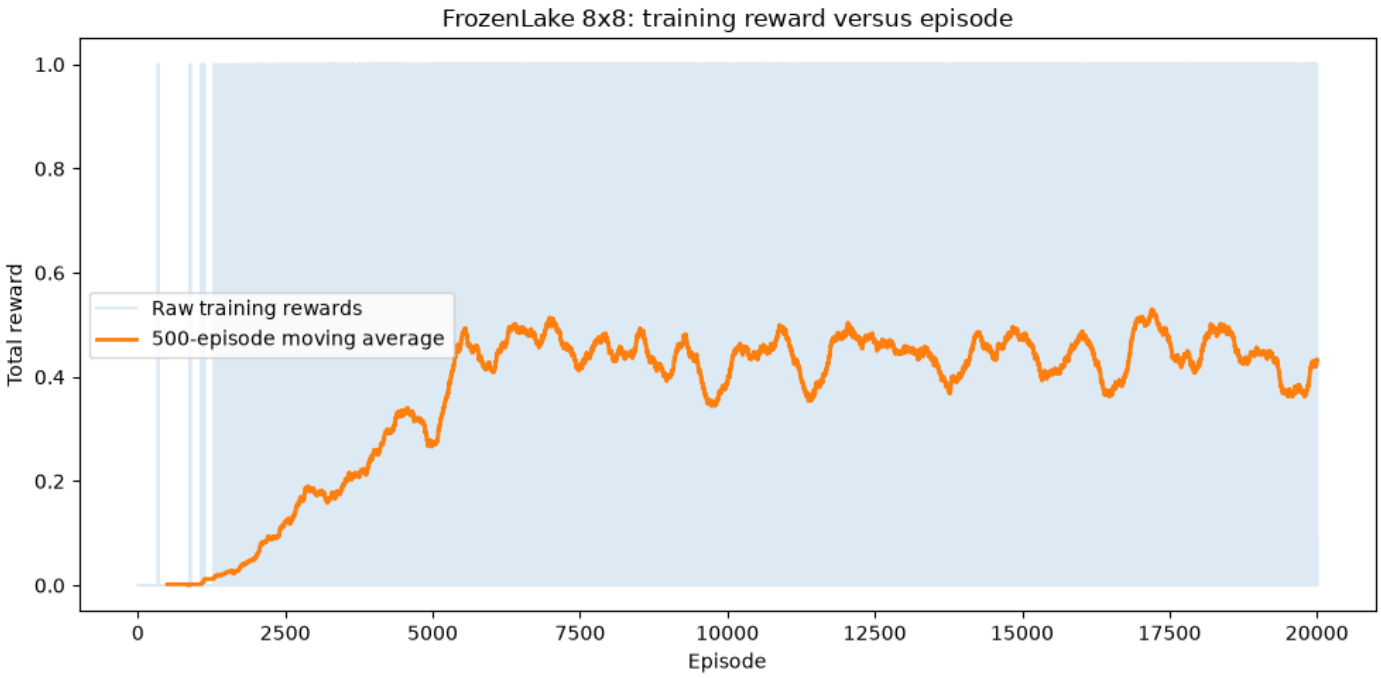


Figure 1. Baseline training rewards and a 500-episode moving average. These rewards include exploratory actions.

The saved baseline finishes with epsilon 0.05, 201 nonzero Q-values, and a highest Q-value of about 0.927. Its mean reward over the last 1,000 training episodes is 0.398, equivalent to a 39.8% training success rate. The moving average rises after the agent discovers useful routes, but continues to fluctuate because both exploratory actions and slippery transitions can change the outcome.

This initial result shows learning, while also showing that the agent still fails often. It is a training measurement, not a separate greedy-policy evaluation. Parts D through F compare settings using the same 20,000-episode budget.

Run provenance: Parts C and D share the original saved baseline. Parts E and F train fresh copies of their baseline-equivalent settings. These are separate stochastic runs, so their baseline-equivalent results need not be identical.

D. Learning-rate experiment

Compare alpha = 0.1, 0.5, and 0.9. Keep gamma, the environment, the exploration schedule, and the episode count fixed at the baseline settings. Alpha 0.1 reuses Part C; the other settings start with fresh Q-tables.

Alpha	First 5,000 mean reward	Last 1,000 mean reward
0.1	0.144	0.398
0.5	0.089	0.278
0.9	0.028	0.124

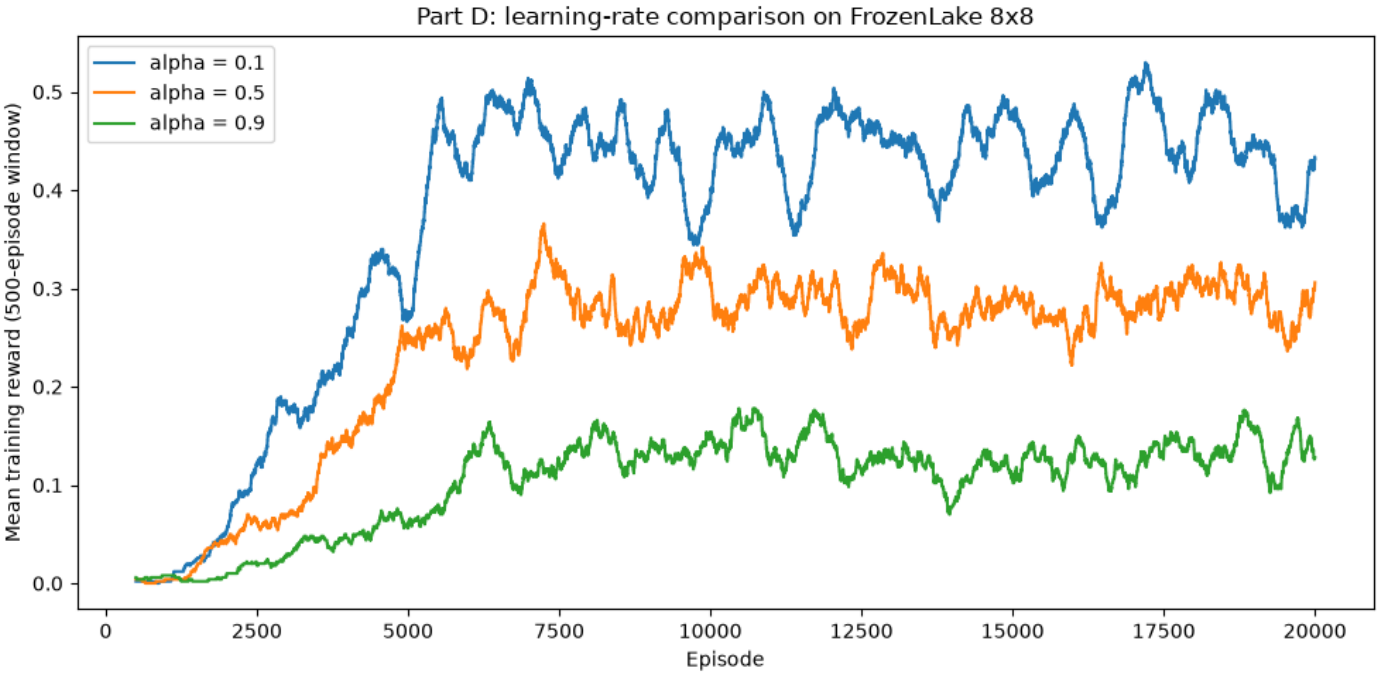


Figure 2. Learning-rate comparison using 500-episode moving averages of training reward.

Alpha 0.1 improves earlier and has the highest early mean reward, 0.144. Larger learning rates do not learn faster in this run. All curves fluctuate, but alpha 0.1 maintains stronger performance after the initial learning period. The final training success rates are 39.8%, 27.8%, and 12.4% for alpha 0.1, 0.5, and 0.9.

A smaller alpha gives less weight to each new sample. This may help when slippery movement produces variable outcomes for the same intended action. The results support choosing alpha 0.1 for the later comparisons. They come from single runs and do not prove that the same ranking will hold under every training seed.

E. Discount-factor experiment

Compare gamma = 0.50, 0.90, and 0.99 while keeping alpha 0.1 and the other baseline settings fixed. All three policies are trained from scratch, including the baseline-equivalent gamma 0.99 setting.

Gamma	First 5,000 mean reward	Last 1,000 mean reward
0.50	0.020	0.037
0.90	0.108	0.299
0.99	0.149	0.471

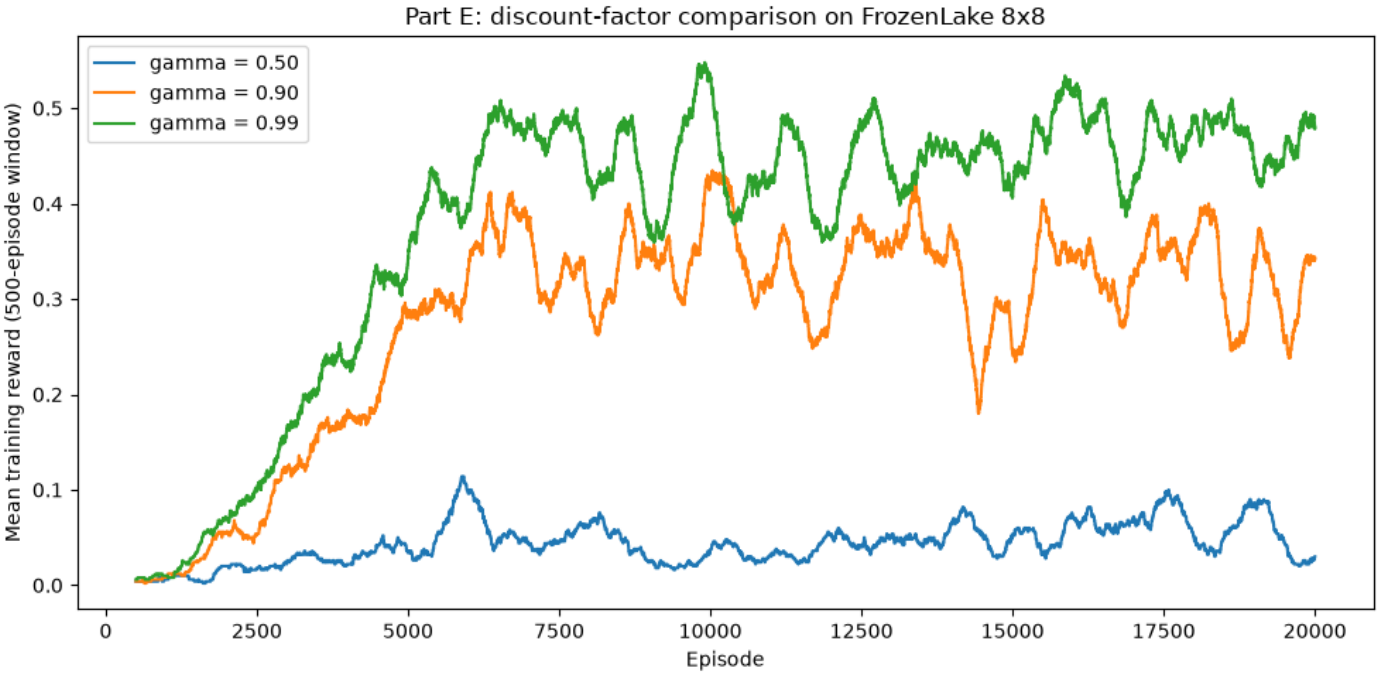


Figure 3. Discount-factor comparison using the same 500-episode training-reward average.

Gamma 0.99 has the highest early mean reward, 0.149, compared with 0.108 and 0.020 for gamma 0.90 and 0.50. All curves fluctuate. After the initial learning period, gamma 0.99 generally maintains the highest rewards. The last-1,000 training success rates are 3.7%, 29.9%, and 47.1% for gamma 0.50, 0.90, and 0.99.

A low discount factor gives much less value to a reward many steps away. FrozenLake gives no positive reward until the goal, so gamma 0.50 weakens the learning signal at earlier states. Gamma 0.99 preserves more of the distant goal reward. This is consistent with its stronger results and motivates keeping gamma 0.99.

F. Exploration-decay experiment

Compare decay factors 0.99, 0.9995, and 0.9999. Keep alpha 0.1, gamma 0.99, the initial and minimum epsilon, and the episode count fixed. Each setting starts a fresh training run.

Decay factor	First 5,000 mean reward	Last 1,000 mean reward	Final epsilon
0.99	0.078	0.459	0.050
0.9995	0.133	0.416	0.050
0.9999	0.009	0.313	0.135

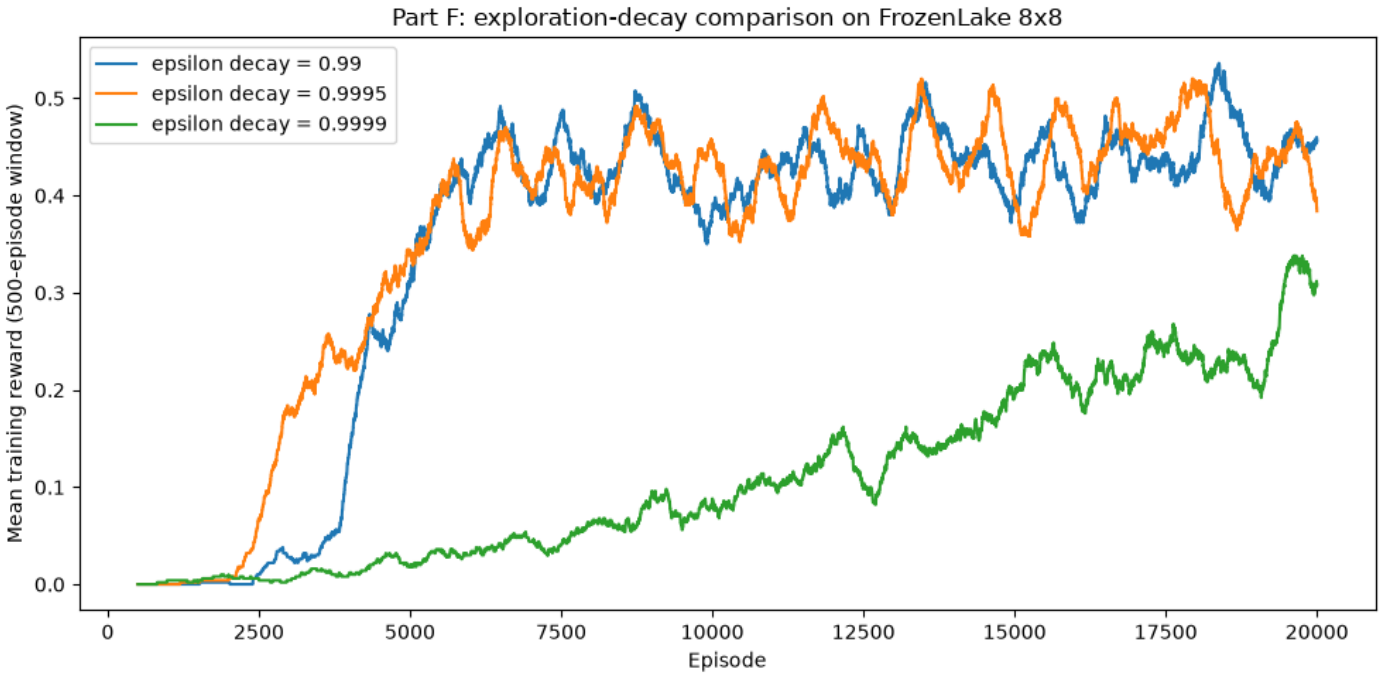


Figure 4. Exploration-decay comparison. Factors closer to 1 retain exploratory actions longer.

The baseline decay 0.9995 improves earlier and has the highest first-5,000 mean reward, 0.133. Decay 0.99 starts more slowly but finishes with the highest last-1,000 mean reward, 0.459. The two curves often reach similar late-training reward levels and both fluctuate. Decay 0.9999 improves more gradually and finishes at 0.313.

Decay 0.99 reaches epsilon 0.05 after 299 episodes; decay 0.9995 reaches it after 5,990 episodes. Decay 0.9999 still has epsilon about 0.135 after 20,000 episodes, so it continues choosing random actions more often. Faster decay uses current preferred actions sooner, but can reduce opportunities to discover better actions. The final training result selects decay 0.99 for Part G; evaluation must check whether that choice improves the greedy policy.

G. Greedy-policy evaluation

Select alpha 0.1, gamma 0.99, and decay 0.99 from the training experiments before looking at evaluation results. Train this configuration and a fresh baseline configuration with seed 42 for 20,000 episodes each. The baseline differs only in decay, which remains 0.9995. Evaluate each policy for 100 episodes on seeds 10000 through 10099, with epsilon zero and no Q-value updates. Greedy ties use the first maximum; slippery movement remains enabled.

Policy	Mean reward	Success rate	Mean steps on success
Baseline	0.370	37.0%	63.2
Selected agent	0.230	23.0%	69.3

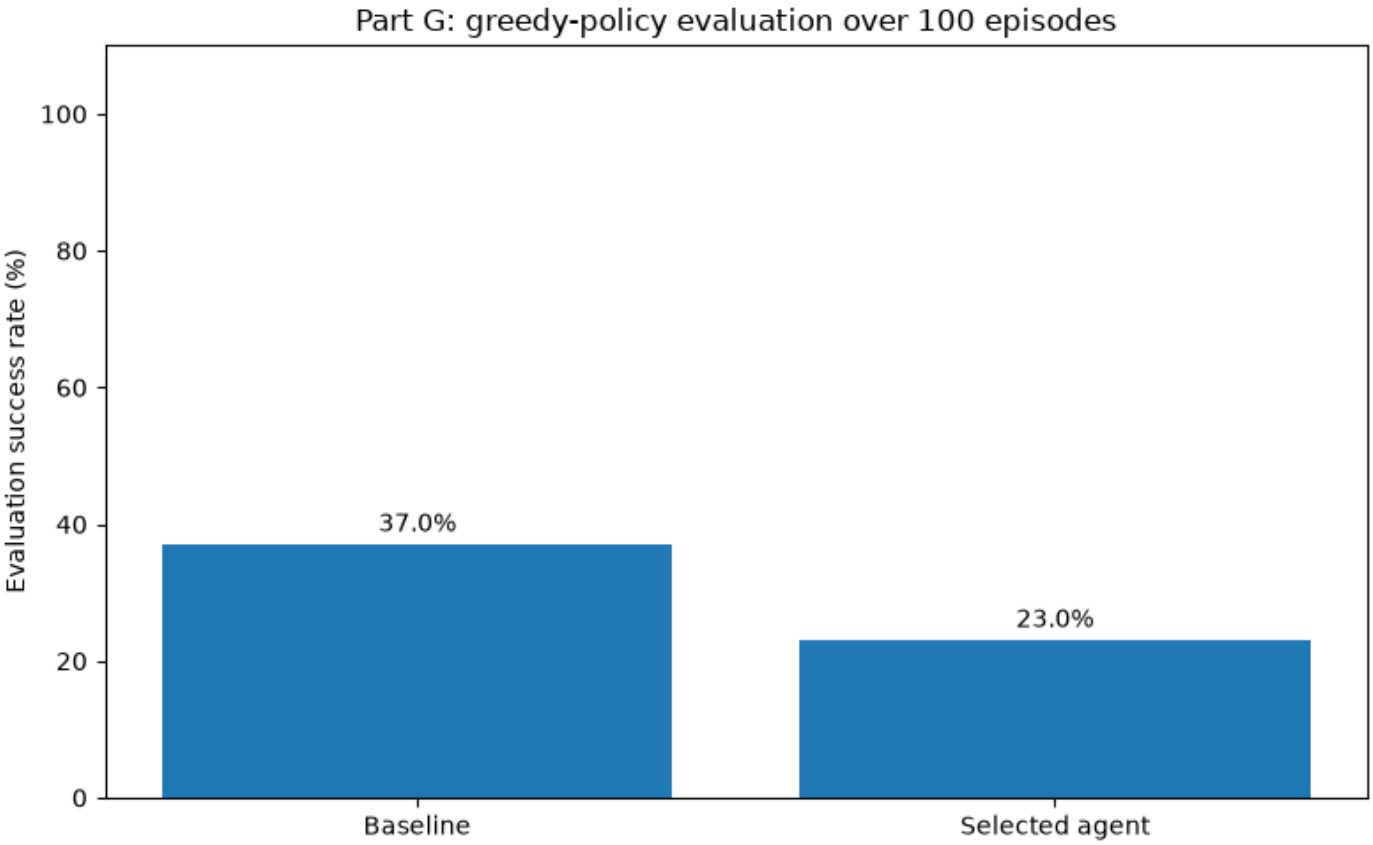


Figure 5. Evaluation success rates. Each policy is tested on the same 100 episode seeds.

The baseline reaches the goal in 37 episodes, compared with 23 for the selected configuration. Mean reward equals success rate because each successful episode earns 1 and every failed episode earns 0. The baseline also takes fewer steps on its successful episodes, although the successful episodes can differ between policies.

In these fresh training runs, the last-1,000 mean rewards are 0.424 for the baseline and 0.437 for the selected agent. The slightly better training score does not carry over to greedy evaluation. This comparison favors retaining the baseline policy, which is used in the video and Part H. The result is from one training seed and 100 evaluation episodes, so it does not establish a universal ranking.

H. Added action noise

Add a separate action-replacement probability p = 0, 0.10, or 0.25. A replacement chooses uniformly among the other three actions before normal slippery movement occurs. Train for 20,000 episodes at each p using the baseline settings and seed 42. Evaluate each matched policy for 100 episodes, then test the unchanged p = 0 policy at every noise level using the same evaluation seeds.

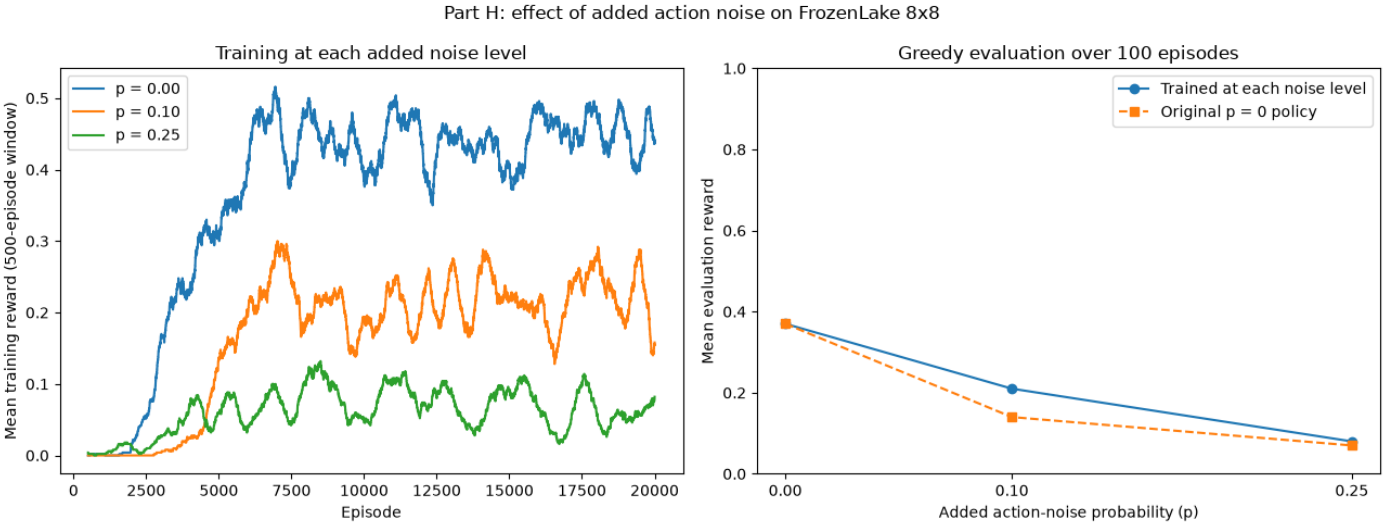


Figure 6. Left: training-reward moving averages. Right: matched-noise and original-policy evaluation. Here p = 0 still includes slippery movement.

Added noise p	First 5,000 mean reward	Last 1,000 mean reward
0.00	0.131	0.424
0.10	0.021	0.220
0.25	0.028	0.069

Added noise p	Matched mean reward	Matched success	Original mean reward	Original success
0.00	0.370	37%	0.370	37%
0.10	0.210	21%	0.140	14%
0.25	0.080	8%	0.070	7%

Both noisy runs have lower early rewards than p = 0, although their early means are not strictly ordered by noise level. The curves fluctuate, and the final training reward falls as noise increases. Matched-noise evaluation success also falls from 37% to 21% and 8%.

The unchanged original policy falls from 37% success to 14% and 7%. Training with matching noise improves success by 7 percentage points at p = 0.10 and 1 point at p = 0.25. The one-episode difference at the highest noise level is too small to show a reliable advantage from this run alone. Checks verify valid different replacement actions, unchanged evaluation Q-tables, and exact original training behavior when p = 0.